Find the Duplicate Number (easy)

Problem Statement

We are given an unsorted array containing `n+1` numbers taken from the range `1` to `n`. The array has only one duplicate but it can be repeated multiple times. **Find that duplicate number without using any extra space.** You are, however, allowed to modify the input array.

Example 1:

```
Input: [1, 4, 4, 3, 2]
Output: 4
```

Example 2:

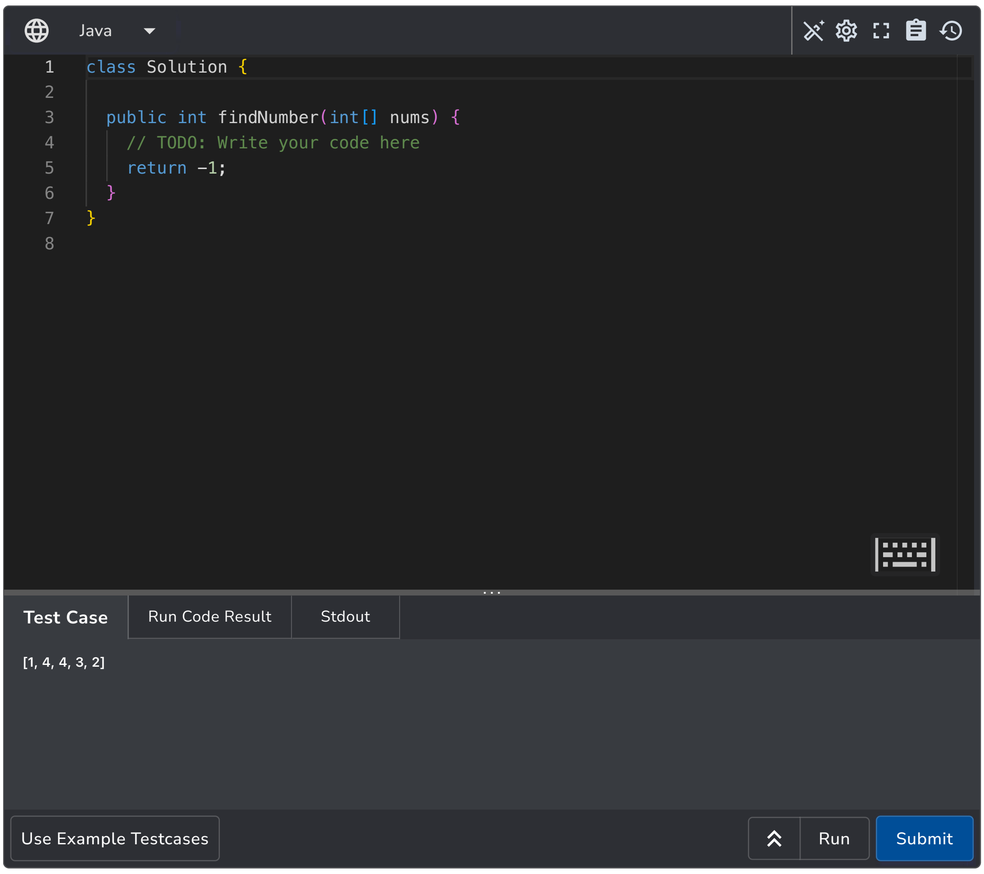
```
Input: [2, 1, 3, 3, 5, 4]
Output: 3
```

Example 3:

```
Input: [2, 4, 1, 4, 4]
Output: 4
```

Try it yourself

Try solving this question here:



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